Chapter 16

The Quantized Electromagnetic Field

16.1 Classical Source-Free Electromagnetism

Throughout these notes, we take $\epsilon_0 = \mu_0 = c = 1$. Consequently, electric and magnetic fields have the same units, the *vacuum dispersion relation* is $\omega_{\mathbf{k}} = |\mathbf{k}|$, and the classical energy is

$$H_{\rm cl} = \frac{1}{2}\int_{\mathbb{R}^3} d^3r \, \left(\mathbf{E}^2 + \mathbf{B}^2\right). \tag{16.1}$$

We will retain $\hbar$, since it identifies the genuinely quantum statements. In vacuum, and in the absence of charges and currents,

$$\nabla\cdot\mathbf{E} = 0, \qquad \nabla\cdot\mathbf{B} = 0, \tag{16.2}$$

$$\nabla\times\mathbf{E} = -\partial_t\mathbf{B}, \qquad \nabla\times\mathbf{B} = \partial_t\mathbf{E}. \tag{16.3}$$

These are the source-free Maxwell equations. With suitable initial data and appropriate boundary conditions, they determine the subsequent evolution of the electromagnetic field.

16.1.1 The Wave Equation

Take the curl of Faraday's law:

$$\nabla\times(\nabla\times\mathbf{E}) = -\partial_t(\nabla\times\mathbf{B}) = -\partial_t^2\mathbf{E}.$$

Using

$$\nabla\times(\nabla\times\mathbf{E}) = \nabla(\nabla\cdot\mathbf{E}) - \nabla^2\mathbf{E} = -\nabla^2\mathbf{E},$$

where the last equality uses $\nabla\cdot\mathbf{E} = 0$, we find

$$(\partial_t^2 - \nabla^2)\mathbf{E} = 0. \tag{16.4}$$

The same calculation with $\mathbf{B}$ gives

$$(\partial_t^2 - \nabla^2)\mathbf{B} = 0. \tag{16.5}$$

The *wave equations* are necessary consequences of Maxwell's equations. They are not by themselves sufficient: a solution of the vector wave equation must also satisfy the divergence constraints and the first-order curl equations.

The general solutions to Eqs. (16.4) & (16.5) are well known and correspond to *linear combinations of traveling plane waves*. We can write a particular solution to Eq. (16.4) as

$$\mathbf{E}(\mathbf{r},t) = \mathbf{C}_\mathbf{k}\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \mathbf{C}_\mathbf{k}^*\, e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} = 2\,\mathrm{Re}\left[\mathbf{C}_\mathbf{k}\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)}\right] \tag{16.6}$$

where $\mathbf{C}_\mathbf{k}$ is a complex amplitude vector, $\mathbf{k}$ is the wavevector, and $\omega > 0$ is the angular frequency of the wave. The complex amplitude vector encodes the real amplitude, direction of the electric field (the polarization), and the phase constant of the wave, while the wavevector describes the direction of wave propagation. In order for Eq. (16.6) to be a valid solution to the wave equation, we must have $\omega_\mathbf{k} = |\mathbf{k}|$. To construct the most general solution to the wave equation, we must combine all the different plane-wave solutions. In general, this superposition takes the form of an integral over all possible values of $\mathbf{k}$. We can simplify the situation by restricting the field to a finite cubic region and imposing periodic boundary conditions. This is merely a mathematical convenience. Once the construction is complete, we can take the limit $L \to \infty$ and recover the continuum description. Under periodic boundary conditions, we require

$$\mathbf{E}(\mathbf{r} + L\,\hat{\mathbf{r}}_j, t) = \mathbf{E}(\mathbf{r}, t) \tag{16.7}$$

where r_j is a Cartesian coordinate. A plane wave must therefore satisfy

$$e^{i\mathbf{k}\cdot(\mathbf{r}+L\hat{\mathbf{r}}_j)} = e^{i\mathbf{k}\cdot\mathbf{r}}, \quad \text{or} \quad e^{ik_jL} = 1, \tag{16.8}$$

so the allowed components of the wavevector are given by $k_jL = 2\pi m_j$, where $m_j \in \mathbb{Z}$. The allowed wavevectors form a discrete lattice in wavevector space.

The most general solution to the wave equation in a finite periodic box can be written as

$$\mathbf{E}(\mathbf{r},t) = \sum_\mathbf{k} \mathbf{C}_\mathbf{k}\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{c.c.}, \tag{16.9}$$

where c.c. denotes the complex conjugate of the first term in the equation. Applying Gauss's law for the electric field,

$$0 = \nabla\cdot\mathbf{E} = i\sum_\mathbf{k} (\mathbf{k}\cdot\mathbf{C}_\mathbf{k})\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{c.c.} \tag{16.10}$$

and it follows that $\mathbf{k}\cdot\mathbf{C}_\mathbf{k} = 0$. Hence, $\mathbf{C}_\mathbf{k}$ lies in a two dimensional plane orthogonal to $\mathbf{k}$. For each

$\mathbf{k} \neq 0$, we choose two real, orthonormal polarization vectors $\{\hat{\mathbf{e}}_{\mathbf{k}\mu}\}$, $\mu = 1, 2$, such that

$$\mathbf{C}_{\mathbf{k}} = \sum_{\mu=1}^{2} C_{\mathbf{k}\mu}\, \hat{\mathbf{e}}_{\mathbf{k}\mu} \tag{16.11}$$

In many applications, it is convenient to choose the set $\left\{\hat{\mathbf{e}}_{\mathbf{k}1}, \hat{\mathbf{e}}_{\mathbf{k}2}, \hat{\mathbf{k}}\right\}$ as a right-handed set of orthogonal unit vectors. Thus, for each nonzero wavevector $\mathbf{k}$, the electric field has two independent transverse polarization components. The complex scalar amplitudes $C_{\mathbf{k}\mu}$ completely specify the amplitude, phase, and polarization of the corresponding plane-wave mode. We may therefore label the classical modes of the free electromagnetic field by the pair $(\mathbf{k}, \mu)$, $\mu = 1, 2$ with angular frequency $\omega_{\mathbf{k}} = |\mathbf{k}|$. It will be convenient to label the modes with one index, i.e., $\lambda = (\mathbf{k}, \mu)$. The electric field can now be written as

$$\mathbf{E}(\mathbf{r}, t) = \sum_{\lambda} C_{\lambda}\, \hat{\mathbf{e}}_{\lambda}\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.12}$$

Starting with Eq. (16.5), we can apply a similar line of reasoning to derive an expression for the magnetic field as a superposition of plane-wave modes:

$$\mathbf{B}(\mathbf{r}, t) = \sum_{\lambda} D_{\lambda}\, \hat{\mathbf{b}}_{\lambda}\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.13}$$

From $\nabla \cdot \mathbf{B} = 0$, it follows that $\mathbf{k} \cdot \hat{\mathbf{b}}_{\lambda} = 0$. Applying Faraday's law,

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \tag{16.14}$$

we find that

$$i \sum_{\lambda} C_{\lambda}\, (\mathbf{k} \times \hat{\mathbf{e}}_{\lambda})\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} = i \sum_{\lambda} D_{\lambda}\, \hat{\mathbf{b}}_{\lambda}\, \omega_{\mathbf{k}} e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.15}$$

Equating coefficients,

$$C_{\lambda}\, (\mathbf{k} \times \hat{\mathbf{e}}_{\lambda}) = D_{\lambda}\, \hat{\mathbf{b}}_{\lambda}\, \omega_{\mathbf{k}} \tag{16.16}$$

Taking the magnitude of both sides, noting that $\hat{\mathbf{e}}_{\lambda}$ is perpendicular to $\mathbf{k}$ and $\omega_{\mathbf{k}} = |\mathbf{k}|$, we find $C_{\lambda} = D_{\lambda}$, and furthermore,

$$\hat{\mathbf{b}}_{\lambda} = \hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\lambda} \tag{16.17}$$

Hence, the magnetic field is

$$\mathbf{B}(\mathbf{r}, t) = \sum_{\lambda} C_{\lambda}\, \left(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\lambda}\right)\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.18}$$

It happens that Eqs. (16.12) & (16.18) are the most general solutions to Maxwell's equations in a finite periodic box. Applying the Ampère-Maxwell equation only tells us that $\omega_{\mathbf{k}} = |\mathbf{k}|$, which we

found by solving the wave equation.

16.1.2 Field Energy

We now wish to compute the field energy in terms of the solutions for $\mathbf{E}$ and $\mathbf{B}$. For each mode, define

$$\phi_{\mathbf{k}}(\mathbf{r}, t) = \mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k}} t,$$

and write the fields as

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*, \qquad \mathbf{E}_+ = \sum_\lambda C_\lambda \hat{\mathbf{e}}_\lambda e^{i\phi_{\mathbf{k}}},$$

$$\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*, \qquad \mathbf{B}_+ = \sum_\lambda C_\lambda \hat{\mathbf{b}}_\lambda e^{i\phi_{\mathbf{k}}}.$$

According to Eq. (16.1), for the total energy, we must integrate $\mathbf{E}^2$ and $\mathbf{B}^2$ over the cube. First consider the electric-field contribution to the energy. Expanding,

$$\mathbf{E}^2 = \mathbf{E}_+ \cdot \mathbf{E}_+ + \mathbf{E}_+^* \cdot \mathbf{E}_+ + \mathbf{E}_+ \cdot \mathbf{E}_+^* + \mathbf{E}_+^* \cdot \mathbf{E}_+^*$$

Isolating one of the mixed terms, we find

$$\int_V \mathbf{E}_+ \cdot \mathbf{E}_+^* \, d^3r = \sum_{\mathbf{k},\mu} \sum_{\mathbf{k}',\nu} C_{\mathbf{k}\mu} C_{\mathbf{k}'\nu}^* \left(\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{\mathbf{k}'\nu}\right) e^{-i(\omega_k - \omega_{k'})t} \int_V e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \, d^3r$$

For a periodic box, we have the following orthogonality condition:

$$\int_V e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \, d^3r = V \delta_{\mathbf{k},\mathbf{k}'}.$$

Hence,

$$\int_V \mathbf{E}_+ \cdot \mathbf{E}_+^* \, d^3r = V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2$$

The other mixed term is just the complex conjugate of the expression above and gives the same result. Together they contribute $2V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2$. As for the unmixed terms, define

$$F = \sum_{\mathbf{k},\mu} \sum_{\mathbf{k}',\nu} C_{\mathbf{k}\mu} C_{\mathbf{k}'\nu} \left(\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{\mathbf{k}'\nu}\right) e^{-i(\omega_k + \omega_{k'})t} \int_V e^{i(\mathbf{k}+\mathbf{k}')\cdot\mathbf{r}} \, d^3r$$

such that

$$\int_V \mathbf{E}^2 \, d^3r = V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2 + F + F^*.$$

The orthogonality relation selects $\mathbf{k}' = -\mathbf{k}$, and since $\omega_{-\mathbf{k}} = \omega_{\mathbf{k}}$, we find

$$F = V \sum_{\mathbf{k},\mu,\nu} C_{\mathbf{k}\mu} C_{-\mathbf{k}\nu} \left(\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{-\mathbf{k}\nu}\right) e^{-2i\omega_k t}$$

By a similar line of reasoning, we arrive at an equivalent expression for the magnetic-field contribution to the field energy:

$$\int_V \mathbf{B}^2 \, d^3r = V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2 + G + G^*,$$

where

$$G = V \sum_{\mathbf{k},\mu,\nu} C_{\mathbf{k}\mu} C_{-\mathbf{k}\nu} \left(\hat{\mathbf{b}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{b}}_{-\mathbf{k}\nu} \right) e^{-i2\omega_k t}$$

Since $\hat{\mathbf{b}}_{\mathbf{k}\mu} = \hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu}$, using the vector identity $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$,

$$\hat{\mathbf{b}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{b}}_{-\mathbf{k}\nu} = \left(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu} \right) \cdot \left(-\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{-\mathbf{k}\nu} \right) = -\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{-\mathbf{k}\nu}$$

Therefore, $G = -F$, and the unmixed terms cancel in the total energy:

$$H_{\text{cl}} = 2V \sum_{\lambda} |C_\lambda|^2 \tag{16.19}$$

Thus, although the fields are constructed by superposing many plane waves, the energy contains no coupling between distinct modes. It is a sum of nonnegative contributions, one from each mode $(\mathbf{k}, \mu)$. In this sense, the normal modes of the free electromagnetic field are *energetically independent*.

16.1.3 Oscillator Form

Since the modes are energetically independent, let

$$H_{\text{cl}} = \sum_{\lambda} H_\lambda, \quad H_\lambda = 2V|C_\lambda|^2 \tag{16.20}$$

Define the time-dependent mode amplitude

$$C_\lambda(t) = C_\lambda e^{-i\omega_{\mathbf{k}} t} \tag{16.21}$$

It is complex, so it contains two real quantities. Define the following change of variables:

$$\begin{aligned} Q_\lambda(t) &= \frac{\sqrt{V}}{i\omega_{\mathbf{k}}} \left[C_\lambda(t) - C_\lambda^*(t) \right] \\ P_\lambda(t) &= -\sqrt{V} \left[C_\lambda(t) + C_\lambda^*(t) \right] \end{aligned} \tag{16.22}$$

Inverting the expressions above, we find

$$C_\lambda(t) + C_\lambda^*(t) = -\frac{P_\lambda(t)}{\sqrt{V}}, \qquad C_\lambda(t) - C_\lambda^*(t) = \frac{i\omega_{\mathbf{k}} Q_\lambda(t)}{\sqrt{V}},$$

and it follows that

$$C_\lambda(t) = \frac{-P_\lambda(t) + i\omega_{\mathbf{k}} Q_\lambda(t)}{2\sqrt{V}}. \tag{16.23}$$

Therefore,

$$|C_\lambda|^2 = \frac{P_\lambda(t)^2 + \omega_{\mathbf{k}}^2 Q_\lambda(t)^2}{4V} \tag{16.24}$$

Finally, since $|C_\lambda(t)| = |C_\lambda|$,

$$H_\lambda = 2V|C_\lambda|^2 = \frac{1}{2}\left(P_\lambda^2 + \omega_{\mathbf{k}}^2 Q_\lambda^2\right). \tag{16.25}$$

While the expression above may have the *mathematical form* of a simple harmonic oscillator, it remains to be shown that the variables Q_λ and P_λ behave dynamically as expected. From Eqs. (16.21) & (16.22), we find

$$\dot{Q}_\lambda = P_\lambda, \quad \text{and} \quad \dot{P}_\lambda = -\omega_{\mathbf{k}}^2 Q_\lambda, \tag{16.26}$$

which are precisely the equations of motion for a simple harmonic oscillator of mass $m = 1$. At this stage, we have shown that the source-free electromagnetic field can be decomposed into a collection of independent simple harmonic oscillators, one for each mode.

16.2 Quantizing the Field

To construct a quantum description that preserves the Maxwell dynamics, we retain the mode expansion but "promote" the real mode variables Q_λ and P_λ to Hermitian operators. We require their Heisenberg equations, generated by the oscillator Hamiltonian, to reproduce the classical equations of motion. Because the electric and magnetic fields are linear combinations of these mode variables, this correspondence ensures that the resulting operator fields obey the source-free Maxwell equations.

Since each oscillator is independent, we consider the Heisenberg evolution of just one mode. The Heisenberg equation for Q is

$$\dot{Q} = \frac{1}{i\hbar}[Q, H], \tag{16.27}$$

where we will drop the λ subscript for now. Since $H = \frac{1}{2}(P^2 + \omega^2 Q^2)$,

$$\begin{aligned}\dot{Q} &= \frac{1}{2i\hbar}[Q, P^2] \\ &= \frac{1}{2i\hbar}\big([Q, P]P + P[Q, P]\big)\end{aligned}$$

Letting $K = [Q, P]$, in order to reproduce Eq. (16.26), we require

$$\{K, P\} = 2i\hbar P, \tag{16.28}$$

where $\{K, P\} = KP + PK$ denotes the *anticommutator* of K and P. Similarly, the Heisenberg equation for P is

$$\dot{P} = \frac{1}{i\hbar}[P, H] = -\frac{\omega^2}{2i\hbar}\{K, Q\}, \tag{16.29}$$

so we require

$$\{K, Q\} = 2i\hbar Q \tag{16.30}$$

The simplest way to satisfy Eqs. (16.28) & (16.30) is to set $K = i\hbar$, where we omit the identity. While this is precisely the choice that leads to a correct description of the quantized electromagnetic field, it is worth mentioning that it is not the most general solution if our only constraint is to satisfy the Maxwell equations.

Consider $K = i\hbar(1+G)$, where G is any linear operator satisfying $\{G, P\} = \{G, Q\} = 0$. Because Q and P are Hermitian, their commutator is anti-Hermitian. So $K^\dagger = -K$, and consequently, $G^\dagger = G$. Furthermore, since G anticommutes with Q and P, it commutes with their squares:

$$[G, Q^2] = [G, P^2] = 0$$

This implies that G commutes with the Hamiltonian and is therefore conserved. Hence,

$$[Q, P] = i\hbar + i\hbar G, \quad \text{where} \quad \{G, P\} = \{G, Q\} = 0, \tag{16.31}$$

and G is a conserved quantity. At this stage, the general solution includes $G \neq 0$.

16.2.1 Creation and Annihilation Operators

When we studied the quantum harmonic oscillator in Chapter 7, we introduced *ladder operators* and used them to determine the oscillator energies and energy eigenstates. Define

$$a = \sqrt{\frac{\hbar\omega}{2\hbar}}Q + \frac{i}{\sqrt{2m\omega}}P, \qquad a^\dagger = \sqrt{\frac{\hbar\omega}{2\hbar}}Q - \frac{i}{\sqrt{2m\omega}}P, \tag{16.32}$$

A direct calculation gives

$$[a, a^\dagger] = \frac{1}{i\hbar}[Q, P] = 1 + G, \tag{16.33}$$

and the anticommutation conditions imply $\{G, a\} = \{G, a^\dagger\} = 0$. Furthermore, a and $a^\dagger$ satisfy

$$\dot{a} = -i\omega a, \quad \dot{a}^\dagger = +i\omega a^\dagger \tag{16.34}$$

as Heisenberg operators. In terms of the ladder operators, the Hamiltonian becomes

$$H = \frac{\hbar\omega}{2}\left(aa^\dagger + a^\dagger a\right) = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right) + \frac{\hbar\omega}{2}G \tag{16.35}$$

We are now in a good place to further constrain the quantized field. Photon-counting experi-

ments show that excitations of a mode are detected as discrete quanta, each with energy $\hbar\omega$, and that a mode can contain any nonnegative integer number of photons, consistent with their bosonic statistics. We therefore choose the standard bosonic representation, in which $a^\dagger a$ is the photon-number operator and $a^\dagger$ raises that number by one while a lowers it by one. As we will show, this representation has $[a, a^\dagger] = 1$; comparing with the generalized relation gives $G = 0$. The Maxwell equations supply the oscillator dynamics while identifying the mode operators with the observed photon-counting states supplies the additional physical input that is needed. In other words, we choose $G = 0$ because photons are observable and obey bosonic statistics.

In summary, each mode of the quantized electromagnetic field behaves like a simple harmonic oscillator with quantized excitations that we call *photons*. The commutation relation $[a, a^\dagger] = 1$ encodes the bosonic nature of photons in a particular mode, and the energy eigenstates indicate the number of photons occupying the mode. Since the modes are independent, the full set of bosonic commutation relations, with $\lambda = (\mathbf{k}, \mu)$, is

$$\boxed{\left[a_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}'\mu'}\right] = \delta_{\mathbf{k}\mathbf{k}'}\delta_{\mu\mu'},} \qquad \boxed{\left[a_{\mathbf{k}\mu}, a_{\mathbf{k}'\mu'}\right] = \left[a^\dagger_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}'\mu'}\right] = 0.} \tag{16.36}$$

The total field Hamiltonian is

$$\boxed{H = \sum_\lambda \hbar\omega_{\mathbf{k}} \left(N_\lambda + \frac{1}{2}\right),} \tag{16.37}$$

where $N_\lambda = a^\dagger_\lambda a_\lambda$ is the number operator for each mode. The lowest energy state of the Hamiltonian is called the **vacuum state** and is denoted $|0\rangle$. The vacuum state satisfies $a_\lambda |0\rangle = 0$ for all modes. The vacuum state is a tensor product of lowest energy states for each mode, i.e.,

$$|0\rangle = |0_{\mathbf{k}_1,\mu_1}\rangle \otimes |0_{\mathbf{k}_2,\mu_2}\rangle \otimes \cdots, \tag{16.38}$$

and the vacuum energy, also called the zero-point energy, is

$$E_{\text{vac}} = \sum_\lambda \frac{\hbar\omega_{\mathbf{k}}}{2} \tag{16.39}$$

Unless there is some cutoff that limits the number of oscillators with arbitrarily high frequencies, this sum diverges because there are an infinite number of modes. Nonetheless, it is convenient to treat E_{vac} as if it were finite. This constant has no effect on the free-field dynamics because it commutes with every operator.

If we apply the creation operator for a particular mode to the vacuum state, we obtain

$$\begin{aligned} a^\dagger_{\mathbf{k}\mu}|0\rangle &= |0_{\mathbf{k}_1,\mu_1}\rangle \otimes |0_{\mathbf{k}_2,\mu_2}\rangle \otimes \cdots a^\dagger_{\mathbf{k}\mu}|0_{\mathbf{k}\mu}\rangle \otimes \cdots \\ &= |0_{\mathbf{k}_1,\mu_1}\rangle \otimes |0_{\mathbf{k}_2,\mu_2}\rangle \otimes \cdots |1_{\mathbf{k}\mu}\rangle \otimes \cdots \end{aligned} \tag{16.40}$$

For simplicity, we denote the single-photon state by $|1_{\mathbf{k}\mu}\rangle$ or $|1_\lambda\rangle$. We denote the state containing n photons in the mode $(\mathbf{k},\mu)$ by

$$|n_{\mathbf{k}\mu}\rangle, \qquad n_{\mathbf{k}\mu} = 0, 1, 2, \ldots. \tag{16.41}$$

This notation describes the occupation of a single mode. A general photon-number state of the entire field must specify the occupation number of every mode. We write such a state as

$$\big|\{n_{\mathbf{k}\mu}\}\big\rangle, \tag{16.42}$$

where the braces denote the complete collection of occupation numbers.

When the ladder operators act on a state of the entire field, they change only the occupation number of the specified mode:

$$a_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle = \sqrt{n_{\mathbf{k}\mu}}\,\big|\ldots, n_{\mathbf{k}\mu}-1, \ldots\big\rangle, \tag{16.43}$$

$$a^\dagger_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle = \sqrt{n_{\mathbf{k}\mu}+1}\,\big|\ldots, n_{\mathbf{k}\mu}+1, \ldots\big\rangle. \tag{16.44}$$

All other occupation numbers remain unchanged. For a single mode with n photons,

$$|n_\lambda\rangle = \frac{\left(a^\dagger_\lambda\right)^n}{\sqrt{n!}}|0\rangle \tag{16.45}$$

and for an arbitrary photon-number state,

$$|\{n_\sigma\}\rangle = \prod_\lambda \frac{\left(a^\dagger_\lambda\right)^{n_\lambda}}{\sqrt{n_\lambda!}}|0\rangle \tag{16.46}$$

Since creation operators for distinct modes commute, the order of the product does not matter. The photon-number states are orthonormal, so an arbitrary field state can be written as

$$|\Psi\rangle = \sum_{\{n_\lambda\}} \psi_{\{n_\lambda\}}\,|\{n_\lambda\}\rangle \tag{16.47}$$

16.2.2 Field Operators

We now turn our attention to the field operators. Begin by writing the fields in terms of (Q_λ, P_λ). Then use Eq. (16.32) to determine the field expansions in terms of a_λ and $a^\dagger_\lambda$. This results in the

same field operators as the direction substitution

$$C_\lambda(t) \longrightarrow i\sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, a_\lambda(t), \qquad C^*_\lambda(t) \longrightarrow -i\sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, a^\dagger_\lambda(t), \tag{16.48}$$

in the classical expressions. Thus, the electric-field operator becomes

$$\boxed{\mathbf{E}(\mathbf{r},t) = i\sum_\lambda \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, \hat{\mathbf{e}}_\lambda \left[a_\lambda e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_{\mathbf{k}}t)} - a^\dagger_\lambda e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_{\mathbf{k}}t)}\right],} \tag{16.49}$$

and the magnetic-field operator is

$$\boxed{\mathbf{B}(\mathbf{r},t) = i\sum_\lambda \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, \hat{\mathbf{b}}_\lambda \left[a_\lambda e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)} - a^\dagger_\lambda e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)}\right].} \tag{16.50}$$

We see that in a quantum-mechanical treatment of the electromagnetic field, the field itself becomes an operator that annihilates and creates photons. Assuming that our procedure has given us the correct expressions for the electric- and magnetic-field operators, we now turn our attention to the genuinely quantum-mechanical features of the quantized electromagnetic field. In a photon-number state, the expectation values of the electric and magnetic fields vanish. This is because each field operator contains creation and annihilation operators, which map a photon-number state to orthogonal states. The expectation values of the squared fields, however, do not vanish. Thus, a photon-number state can possess nonzero electromagnetic energy and *nonzero field fluctuations* even though its mean electric and magnetic fields are zero. The vacuum provides the simplest example: its mean fields vanish, but its mean-square fields are nonzero.

A nonzero mean electric or magnetic field requires a coherent superposition of states with different photon numbers. More precisely, because the field operators change photon number by one, the state must contain coherence between photon-number states whose occupation numbers differ by one. An incoherent mixture of different photon numbers still has a vanishing mean field. **Coherent states** are the most important example of states with nonzero mean fields. Their field expectation values have the same plane-wave form as classical electromagnetic fields, while their photon numbers are not definite. When the mean photon number is large, the relative photon-number fluctuations become small, and the coherent state increasingly resembles a classical wave.

Example 16.1: Photon-Number States

Let $|\{n_{\mathbf{k}\mu}\}\rangle$ denote a photon-number state in a finite periodic box. Calculate the expectation values of $\mathbf{E}$, $\mathbf{B}$, $\mathbf{E}^2$, and $\mathbf{B}^2$. Use the last two expressions to calculate the expectation value of H_{field} and confirm that it gives the correct energy.

Solution. For every mode,

$$\langle\{n\}|a_{\mathbf{k}\mu}|\{n\}\rangle = 0, \qquad \langle\{n\}|a^\dagger_{\mathbf{k}\mu}|\{n\}\rangle = 0\rangle\,,$$

because the ladder operators change the photon occupation number. Therefore,

$$\langle\mathbf{E}(\mathbf{r},t)\rangle = 0, \qquad \langle\mathbf{B}(\mathbf{r},t)\rangle = 0.$$

For products of ladder operators,

$$\left\langle a_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}\right\rangle = \left\langle a^\dagger_{\mathbf{k}\mu}a^\dagger_{\mathbf{k}'\mu'}\right\rangle = 0,$$

while

$$\left\langle a^\dagger_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}\right\rangle = n_{\mathbf{k}\mu}\delta_{\mathbf{k}\mathbf{k}'}\delta_{\mu\mu'}$$

and

$$\left\langle a_{\mathbf{k}\mu}a^\dagger_{\mathbf{k}'\mu'}\right\rangle = (n_{\mathbf{k}\mu}+1)\,\delta_{\mathbf{k}\mathbf{k}'}\delta_{\mu\mu'}.$$

Using $\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{e}}_{\mathbf{k}\mu} = 1$, we obtain

$$\begin{aligned}\langle\mathbf{E}^2(\mathbf{r},t)\rangle &= \sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{2V}\left[\left\langle a_{\mathbf{k}\mu}a^\dagger_{\mathbf{k}\mu}\right\rangle + \left\langle a^\dagger_{\mathbf{k}\mu}a_{\mathbf{k}\mu}\right\rangle\right]\\ &= \sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{2V}\left[(n_{\mathbf{k}\mu}+1)+n_{\mathbf{k}\mu}\right].\end{aligned}$$

Hence,

$$\boxed{\langle\mathbf{E}^2(\mathbf{r},t)\rangle = \sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{2V}\left(2n_{\mathbf{k}\mu}+1\right).}$$

The magnetic field calculation gives the same result, i.e., $\left\langle\mathbf{E}^2\right\rangle = \left\langle\mathbf{B}^2\right\rangle$. The expectation value of the field Hamiltonian is

$$\begin{aligned}\langle H\rangle &= \frac{1}{2}\int_V d^3r\,\left(\langle\mathbf{E}^2\rangle + \langle\mathbf{B}^2\rangle\right)\\ &= \frac{1}{2}V\sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{V}\left(2n_{\mathbf{k}\mu}+1\right)\\ &= \sum_{\mathbf{k},\mu}\hbar\omega_{\mathbf{k}}\left(n_{\mathbf{k}\mu}+\frac{1}{2}\right).\end{aligned}$$

This agrees with the result obtained from computing the expectation value directly using $\langle\{n\}|H|\{n\}\rangle$; it's just the energy eigenvalue of the photon-number state.

Hence, for an electromagnetic field prepared in a photon-number state,

$$\langle \mathbf{E} \rangle = 0, \qquad \langle \mathbf{E}^2 \rangle > 0.$$

The first equation says that the probability distribution for a suitable electric-field measurement is centered at zero. The second says that the distribution has nonzero width. Therefore, individual measurements need not return zero; repeated measurements on identically prepared states produce different outcomes whose average approaches zero.

16.2.3 Field Momentum

Classically, we know that the electromagnetic field carries momentum as well as energy. In units with $c = 1$, the momentum density of the electromagnetic field is

$$\mathbf{g}(\mathbf{r}, t) = \mathbf{E} \times \mathbf{B} \tag{16.51}$$

so the total field momentum is

$$\mathbf{P} = \int d^3r \; \mathbf{E} \times \mathbf{B} \tag{16.52}$$

For a plane wave, the fields are transverse and $\mathbf{B} = \hat{\mathbf{k}} \times \mathbf{E}$, so its momentum points in the direction of propagation $\hat{\mathbf{k}}$. For the quantized field, despite the ordering ambiguity, Eq. (16.52) is the correct expression for the total field momentum operator. As an operator, it represents the *generator of translations of the field itself*. Substituting the mode expansions for $\mathbf{E}$ and $\mathbf{B}$ into Eq. (16.52),

$$\mathbf{P} = \sum_{\mathbf{k},\mu} \hbar \mathbf{k} a^\dagger_{\mathbf{k}\mu} a_{\mathbf{k}\mu} = \sum_{\mathbf{k},\mu} \hbar \mathbf{k} N_{\mathbf{k},\mu}, \tag{16.53}$$

where we've used the fact that $\sum_{\mathbf{k},\mu} \hbar \mathbf{k} = 0$ since for every value of $\mathbf{k}$, there is a $-\mathbf{k}$ to cancel it.

As expected, the vacuum state has no momentum. Applying the momentum operator to a single-photon state,

$$\mathbf{P} \, |1_{\mathbf{k}\mu}\rangle = \sum_{\mathbf{k}',\mu'} \hbar \mathbf{k}' N_{\mathbf{k}'\mu} \, |1_{\mathbf{k}\mu}\rangle = \hbar \mathbf{k} \, |1_{\mathbf{k}\mu}\rangle$$

Evidently, a single photon carries momentum $\hbar\mathbf{k}$ in addition to its energy $\hbar\omega$. Since $\omega = |\mathbf{k}|$, we find that the energy of a photon is related to its momentum: $E = P$. If we interpret E and P as the relativistic energy and momentum, this would imply that photons are massless according to the energy-momentum relation from special relativity.

16.2.4 Angular Momentum

The classical field also carries angular momentum. For a localized field configuration, its total angular momentum about the origin is

$$\mathbf{J} = \int d^3r\, \mathbf{r} \times \big(\mathbf{E} \times \mathbf{B}\big) \tag{16.54}$$

Once again, despite the ordering ambiguity, Eq. (16.54) is the correct expression for the total angular momentum operator. As an operator it represents the *generator of rotations of the field.* A rotation of the electromagnetic field changes both where the field is evaluated and its vector components.

For a single photon, the most direct intrinsic angular-momentum property is its **helicity**, defined as the component of its spin along its direction of motion:

$$\Lambda_{\mathbf{k}} = \hat{\mathbf{k}} \cdot \mathbf{J}. \tag{16.55}$$

For a plane wave, a rotation about $\hat{\mathbf{k}}$ leaves the spatial dependence of the wave unchanged because the field is uniform on the surface of each plane perpendicular to $\hat{\mathbf{k}}$. Its angular momentum about that axis therefore acts on the transverse polarization components. Let $|1_{\mathbf{k}1}\rangle$ and $|1_{\mathbf{k}2}\rangle$ denote one-photon states with the two linear polarizations and define the **circularly polarized states**

$$|1_{\mathbf{k},\sigma}\rangle = \frac{1}{\sqrt{2}}\big(|1_{\mathbf{k}1}\rangle + i\sigma|1_{\mathbf{k}2}\rangle\big), \qquad \sigma = \pm 1. \tag{16.56}$$

$$\Lambda_{\mathbf{k}}|1_{\mathbf{k},\sigma}\rangle = \sigma\hbar|1_{\mathbf{k},\sigma}\rangle. \tag{16.57}$$

Thus, the two allowed helicity eigenvalues are $+\hbar$ and $-\hbar$. The photon has an intrinsic spin of 1 because the electromagnetic field transforms as a vector under rotations. But of course, we would not be speaking of photons if it weren't for the bosonic commutation relations we used to arrive at the photon picture. This is our first glimpse into the spin-statistics theorem!

16.3 Jaynes-Cummings Model

Having constructed a quantum description of the electromagnetic field, we can now ask how the field interacts with matter. In general, this is a complicated problem: an atom has many energy levels, and the field contains many modes, all of which may participate in the interaction. We will develop a broader treatment of light–matter interactions in the next chapter. For now, we will focus on a simple model that isolates some of the essential physics.

The Jaynes–Cummings model describes a two-level atom interacting with a single mode of the quantized electromagnetic field. The two-level approximation retains just two atomic states, while the single-mode approximation keeps just one field oscillator. With the dipole and rotating-wave

approximations, the Hamiltonian can be written as

$$H = \hbar\omega\, a^\dagger a + \frac{\hbar\omega_0}{2}\sigma_z + \hbar g\left(\sigma_+ a + \sigma_- a^\dagger\right), \tag{16.58}$$

where the operators σ_z, σ_+, and σ_- act on the atomic states while the ladder operators $a, a^\dagger$ act on the photon-number states. The parameter ω refers to the frequency of the field, ω_0 is related to the energy separation between the two atomic states, and g describes the strength of the interaction between the atom and the field. Despite its simplicity, this model captures a central feature of light–matter interaction: the atom and field can exchange discrete excitations. It will let us examine that exchange and the resulting atom–field dynamics in a setting simple enough to solve exactly.

16.3.1 Derivation of the Hamiltonian

Consider a two-level atom with ground state $|g\rangle$, excited state $|e\rangle$, and energy separation $\hbar\omega_0$. The free atomic Hamiltonian may be written as

$$H_{\text{atom}} = \frac{\hbar\omega_0}{2}\sigma_z, \tag{16.59}$$

where σ_z is a Pauli operator. We may formulate the atomic Hamiltonian in matrix representation by choosing the standard orthonormal basis for $\{|g\rangle, |e\rangle\}$. More precisely, we set

$$|e\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |g\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \tag{16.60}$$

so that

$$\sigma_z = |e\rangle\langle e| - |g\rangle\langle g| = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}, \tag{16.61}$$

and

$$H_{\text{atom}} = \frac{\hbar\omega_0}{2}\begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}. \tag{16.62}$$

We also define the atomic ladder operators

$$\sigma_+ = |e\rangle\langle g| = \begin{pmatrix}0 & 1\\0 & 0\end{pmatrix}, \quad \sigma_- = |g\rangle\langle e| = \begin{pmatrix}0 & 0\\1 & 0\end{pmatrix}. \tag{16.63}$$

The two-level system is interacting with a single mode of the electromagnetic field characterized by angular frequency ω, wavevector $\mathbf{k}$, and polarization μ. Neglecting the zero-point energy, the free-field Hamiltonian for this mode is

$$H_{\text{field}} = \hbar\omega\, a^\dagger a \tag{16.64}$$

The free Hamiltonian of the combined system, neglecting interactions, is therefore

$$H_0 = \frac{\hbar\omega_0}{2}\,\sigma_z + \hbar\omega\, a^\dagger a \tag{16.65}$$

Assuming the wavelength of the radiation is much larger than the characteristic size of the two-level system, we use the **electric-dipole approximation** to model the interaction Hamiltonian as

$$V = -\mathbf{d}\cdot\mathbf{E}(\mathbf{r}_0) \tag{16.66}$$

here $\mathbf{d}$ is the dipole operator of the two-level system and $\mathbf{r}_0$ is its location. Within the two-level subspace, the dipole operator may be written in matrix representation as

$$\mathbf{d} = \begin{pmatrix} \mathbf{d}_{ee} & \mathbf{d}_{eg} \\ \mathbf{d}_{ge} & \mathbf{d}_{gg} \end{pmatrix} \tag{16.67}$$

where $\mathbf{d}_{ij} = \langle i|\mathbf{d}|j\rangle$ denotes a matrix element of the dipole operator in the atomic states. Many two-atoms lack a permanent dipole moment in their energy eigenstates, so we will set the diagonal terms to zero. Since $\mathbf{d}$ must be Hermitian, $\mathbf{d}_{eg} = \mathbf{d}_{ge}^*$. In operator form,

$$\mathbf{d} = \mathbf{d}_{eg}\sigma_+ + \mathbf{d}_{ge}\sigma_-. \tag{16.68}$$

For the selected mode, the Schrödinger-picture electric-field operator at the position of the two-level atom is

$$\mathbf{E}(\mathbf{r}_0) = i\mathcal{E}\,\hat{\mathbf{e}}\left(a e^{i\mathbf{k}\cdot\mathbf{r}_0} - a^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}_0}\right) = \mathcal{E}\,\hat{\mathbf{e}}\left(a + a^\dagger\right), \tag{16.69}$$

where $\mathcal{E} = \sqrt{\hbar\omega/2V}$, and in the second equality, we've absorbed the constant phases coming from i and $e^{i\mathbf{k}\cdot\mathbf{r}_0}$ into the definitions of the ladder operators. In other words,

$$ia e^{i\mathbf{k}\cdot\mathbf{r}_0} \to a, \qquad -ia^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}_0} \to a^\dagger$$

This redefinition does not affect the bosonic commutation relations, and therefore has no theoretical or empirical consequence.

The interaction Hamiltonian is given by

$$\begin{aligned} V &= -\left(\mathbf{d}_{eg}\sigma_+ + \mathbf{d}_{ge}\sigma_-\right)\cdot\left[\mathcal{E}\hat{\mathbf{e}}(a + a^\dagger)\right] \\ &= -\mathcal{E}\left[\left(\mathbf{d}_{eg}\cdot\hat{\mathbf{e}}\right)\sigma_+ + \left(\mathbf{d}_{ge}\cdot\hat{\mathbf{e}}\right)\sigma_-\right](a + a^\dagger). \end{aligned}$$

The atomic and field operators act on independent vector spaces, so they commute. We may expand the product:

$$V = -\mathcal{E}\Big[\left(\mathbf{d}_{eg}\cdot\hat{\mathbf{e}}\right)\sigma_+ a + \left(\mathbf{d}_{eg}\cdot\hat{\mathbf{e}}\right)\sigma_+ a^\dagger + \left(\mathbf{d}_{ge}\cdot\hat{\mathbf{e}}\right)\sigma_- a + \left(\mathbf{d}_{ge}\cdot\hat{\mathbf{e}}\right)\sigma_- a^\dagger\Big].$$

Define the complex coupling constant g via

$$\hbar g = -\mathcal{E}\left(\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right).$$

For a real polarization vector, $\mathbf{d}_{ge} \cdot \widehat{\mathbf{e}} = \left(\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right)^*$, and futhermore, $\hbar g^* = -\mathcal{E}\left(\mathbf{d}_{ge} \cdot \widehat{\mathbf{e}}\right)$. The interaction Hamiltonian becomes

$$V = \left(\hbar g \sigma_+ + \hbar g^* \sigma_-\right)\left(a + a^\dagger\right)$$

The phase of g can be absorbed into the phases of $|e\rangle$ and $|g\rangle$. More precisely, let $g = |g|e^{i\delta}$ in polar form, where $\delta \in \mathbb{R}$. Making the substitutions $|\tilde{e}\rangle = |e\rangle\, e^{i\delta/2}$, $|\tilde{g}\rangle = |g\rangle\, e^{-i\delta/2}$, we find

$$V = \hbar|g|\left(\,|\tilde{e}\rangle\langle\tilde{g}| + |\tilde{g}\rangle\langle\tilde{e}|\,\right)\left(a + a^\dagger\right)$$

Since the atomic states are only well-defined up to an overall phase, we can simply use a new basis in terms of $|\tilde{e}\rangle$ and $|\tilde{g}\rangle$, and we will subsequently relabel these states $|e\rangle$ and $|g\rangle$. We may therefore choose g to be real and positive. Its magnitude is

$$g = \frac{\mathcal{E}}{\hbar}\left|\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right| = \sqrt{\frac{\omega}{2\hbar V}}\;\left|\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right|.$$

In the end, our interaction Hamiltonian may be written in the form

$$V = \hbar g\left(\sigma_+ a + \sigma_+ a^\dagger + \sigma_- a + \sigma_- a^\dagger\right).$$

Let's study the action of this Hamiltonian on the arbitrary product states $|g\rangle|n\rangle$ and $|e\rangle|n\rangle$, where the first term in the product refers to the atomic state and the second term refers to the photon-number state with $n > 0$. The term $\sigma_+ a$ excites the two-level system while annihilating a photon. It therefore describes *photon absorption*. The term $\sigma_- a^\dagger$ lowers the two-level system while creating a photon, thus describing *photon emission*. The free-energy difference between states involved in photon absorption or emission is $\hbar|\omega - \omega_0|$. which is small when the field is nearly resonant with the transition.

The remaining terms simultaneously excite both systems or simultaneously lower both systems. These terms couple states whose free energies differ by $\hbar(\omega + \omega_0)$. Consequently, the relative phases of the coupled states vary rapidly at frequency $\omega + \omega_0$, causing the effects of these terms to largely average out over the longer time scale of the near-resonant dynamics. When the coupling is weak compared with $\omega + \omega_0$, we therefore neglect these two terms. This is the **rotating-wave approximation**, and we write

$$V \approx V_{\text{RWA}} = \hbar g\left(\sigma_+ a + \sigma_- a^\dagger\right)$$

The rotating wave approximation is valid in the near-resonant, weak-coupling regime. Remind me to provide a better explanation when we cover time-dependent perturbation theory!

Adding everything together, we find Eq. (16.58) in its conventional form. The Jaynes–Cummmings model applies approximately to systems such as atoms in optical or microwave cavities, Rydberg atoms, quantum dots, and superconducting qubits coupled to resonators. Its validity relies on the single-mode, two-level, electric-dipole, and rotating-wave approximations, together with the neglect of dissipation. It captures many of the central features of quantum light–matter interaction, including photon absorption and emission, vacuum Rabi oscillations, and atom–field entanglement.

16.3.2 Excitation Conservation

Let n, the number of photons, be the excitation number of the field. Likewise, the atom has an excitation number of 1 in its excited state and 0 in its ground state. Define the **total excitation number** N as the sum of excitations and the **excitation number operator** as

$$N_{\text{tot}} = a^\dagger a + |e\rangle\langle e| \tag{16.70}$$

As it turns out, the Jaynes-Cummings Hamiltonian commutes with N_{tot}. This means the total excitation number is conserved, and we can search for common eigenstates of N_{tot} and H. Consider the action of the number operator on the state $|g\rangle|n\rangle$ and $|e\rangle|n\rangle$:

$$N_{\text{tot}}|g\rangle|n\rangle = n|g\rangle|n\rangle, \quad N_{\text{tot}}|e\rangle|n\rangle = (n+1)|g\rangle|n\rangle \tag{16.71}$$

Thus, the eigenvalue $N = 0$ has the single state $|g\rangle|0\rangle$. For each $N \geq 1$, the eigenvalue N is twofold degenerate, with an eigenspace spanned by $|g\rangle|N\rangle$ and $|e\rangle|N-1\rangle$. Commutation guarantees that H and N_{tot} share eigenspaces; within each degenerate eigenspace of N_{tot}, we must diagonalize H to find the shared eigenvectors.

Let's construct the Hamiltonian within the eigenspace $\{|e\rangle|n\rangle, |g\rangle|n+1\rangle\}$. The total excitation number in this eigenspace is $n+1$. The free Hamiltonian is already diagonal in this eigenspace; we need to compute the action of the interaction term:

$$V|e\rangle|n\rangle = \hbar g\sqrt{n+1}\,|g\rangle|n+1\rangle \tag{16.72}$$

and

$$V|g\rangle|n+1\rangle = \hbar g\sqrt{n+1}\,|e\rangle|n\rangle \tag{16.73}$$

Evidently, the free Hamiltonian accounts for the diagonal terms while the interaction Hamiltonian accounts for off-diagonal terms. This is easier to see in matrix notation. Let

$$|e\rangle|n\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |g\rangle|n+1\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \tag{16.74}$$

In matrix representation, the Hamiltonian in a fixed-excitation-number eigenspace is

$$H_n = \begin{pmatrix} n\hbar\omega + \hbar\omega_0/2 & \hbar g\sqrt{n+1} \\ \hbar g\sqrt{n+1} & (n+1)\hbar\omega - \hbar\omega_0/2 \end{pmatrix} \tag{16.75}$$

The eigenvalues of H_n are found to be

$$E_\pm = \hbar\omega\left(n + \frac{1}{2}\right) \pm \frac{\hbar}{2}\sqrt{(\omega_0 - \omega)^2 + 4g^2(n+1)}, \tag{16.76}$$

and an orthonormal set of corresponding eigenvectors may be written as

$$\begin{aligned} |n+\rangle &= \cos\frac{\theta_n}{2}\,|e\rangle|n\rangle + \sin\frac{\theta_n}{2}\,|g\rangle|n+1\rangle, \\ |n-\rangle &= \sin\frac{\theta_n}{2}\,|e\rangle|n\rangle - \cos\frac{\theta_n}{2}\,|g\rangle|n+1\rangle. \end{aligned} \tag{16.77}$$

where the angle θ_n is defined through

$$\tan\theta_n = \frac{2g\sqrt{n+1}}{\omega_0 - \omega}. \tag{16.78}$$

16.3.3 Rabi Oscillations

If the field initially contains n photons and the atom is excited, the initial state is

$$|\Psi(0)\rangle = |e\rangle|n\rangle.$$

Since the Jaynes–Cummings Hamiltonian preserves excitation number, the state at any later time is spanned by $|e\rangle\,|n\rangle$ and $|g\rangle\,|n+1\rangle$ as we found in the last section. We write

$$|\Psi(t)\rangle = c_e(t)|e\rangle|n\rangle + c_g(t)|g\rangle|n+1\rangle.$$

Define the **detuning** $\Delta = \omega_0 - \omega$ and the **generalized Rabi frequency** $\Omega_n = \sqrt{\Delta^2 + 4g^2(n+1)}$. Evolving the initial state with the fixed-excitation Hamiltonian gives

$$|c_g(t)|^2 = \frac{4g^2(n+1)}{\Omega_n^2}\sin^2\left(\frac{\Omega_n t}{2}\right). \tag{16.79}$$

This is the probability that the atom has made a transition to its ground state while the field has gained one photon. The probability oscillates with angular frequency Ω_n, since

$$\sin^2\left(\frac{\Omega_n t}{2}\right) = \frac{1 - \cos(\Omega_n t)}{2}.$$

Its period is $2\pi/\Omega_n$. On resonance, $\omega = \omega_0$, it reduces to

$$|c_g(t)|^2 = \sin^2\!\big(gt\sqrt{n+1}\big). \tag{16.80}$$

The atom is fully in its ground state at $t = \pi/(2g\sqrt{n+1})$, and the excitation flows back into the atom. Increasing g increases the oscillation frequency, so the atom and field exchange energy more rapidly. Increasing the initial photon number also speeds up the exchange, through the factor $\sqrt{n+1}$. The n-dependent contribution reflects **stimulated emission**—photons already in the mode enhance the emission process. Stimulated emission is the basic amplification process used in lasers. The formula also applies when $n = 0$ and therefore describes **spontaneous emission**.

Example 16.2: Rabi Oscillations

Let's work through a simpler example and show more details of the calculation. Consider a two-level system resonantly coupled to a single mode of the electromagnetic field, so that $\omega = \omega_0$. Initially, the two-level system is in its excited state and the cavity mode is in the vacuum state:

$$|\psi(0)\rangle = |e, 0\rangle .$$

Determine the state $|\psi(t)\rangle$ at time t. Then find the probability that the system has emitted a photon into the cavity and the earliest time at which this probability reaches its maximum value.

Solution. The Jaynes–Cummings Hamiltonian conserves the total number of excitations. The initial state $|e, 0\rangle$ contains one excitation, so the evolution remains in the one-excitation subspace. Hence,

$$|\psi(t)\rangle = c_e(t)\,|e, 0\rangle + c_g(t)\,|g, 1\rangle .$$

The coefficient $c_e(t)$ is the amplitude for the two-level system to remain excited with no photon in the cavity. The coefficient $c_g(t)$ is the amplitude for the two-level system to be in its ground state with one photon in the cavity.

In the ordered basis $\{|e, 0\rangle , |g, 1\rangle\}$, the Hamiltonian is

$$H = \begin{pmatrix} \langle e,0|H|e,0\rangle & \langle e,0|H|g,1\rangle \\ \langle g,1|H|e,0\rangle & \langle g,1|H|g,1\rangle \end{pmatrix} = \hbar \begin{pmatrix} \omega & g \\ g & \omega \end{pmatrix}.$$

The eigenvectors are easy to find; denote them by

$$|+\rangle = \frac{1}{\sqrt{2}}\left(|e, 0\rangle + |g, 1\rangle\right)$$

and

$$|-\rangle = \frac{1}{\sqrt{2}}\left(|e,0\rangle - |g,1\rangle\right)$$

The corresponding energies are $E_\pm = \hbar(\omega \pm g)$. The time-evolution operator is

$$U(t) = e^{-iHt/\hbar} = \frac{e^{-i(\omega+g)t}}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} + \frac{e^{-i(\omega-g)t}}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}.$$

We separate the common factor of $e^{-i\omega t}$ and simplify using Euler's identity:

$$U(t) = e^{-i\omega t}\begin{pmatrix}\cos(gt) & -i\sin(gt)\\ -i\sin(gt) & \cos(gt)\end{pmatrix}.$$

The complete time-evolved state is

$$|\psi(t)\rangle = e^{-i\omega t}\Big[\cos(gt)\,|e,0\rangle - i\sin(gt)\,|g,1\rangle\Big]$$

and the probability the system has emitted a photon into the cavity is

$$P_{\text{emit}}(t) = |c_g(t)|^2 = \sin^2(gt)$$

The probability is first maximized when $gt = \pi/2$, or

$$t_{\max} = \frac{\pi}{2g}$$

At this time, $|\psi(t_{\max})\rangle = |g,1\rangle$. The excitation has been transferred from the two-level system to the cavity field. Because the ideal cavity is lossless, the photon will subsequently be reabsorbed, and the excitation will continue to oscillate between $|e,0\rangle$ and $|g,1\rangle$.

16.4 Continuum Limit

We now let $V = L^3 \to \infty$. Since the spacing between neighboring components of the wavevector is $\Delta k_j = 2\pi/L$, one allowed wavevector occupies a volume $\Delta^3 k = (2\pi)^3/V$ in wavevector space. In the continuum limit, we replace sums with integrals:

$$\sum_{\mathbf{k}} \longrightarrow \frac{V}{(2\pi)^3}\int d^3k,$$

and we rescale the ladder operators by introducing

$$a_\mu(\mathbf{k}) = \sqrt{\frac{V}{(2\pi)^3}}\; a_{\mathbf{k}\mu} \tag{16.81}$$

In the continuum limit, the discrete commutation relations are replaced by

$$\begin{aligned}\left[a_\mu(\mathbf{k}), a^\dagger_{\mu'}(\mathbf{k}')\right] &= \delta_{\mu\mu'}\delta^{(3)}(\mathbf{k}-\mathbf{k}'),\\ \left[a_\mu(\mathbf{k}), a_{\mu'}(\mathbf{k}')\right] &= \left[a^\dagger_\mu(\mathbf{k}), a^\dagger_{\mu'}(\mathbf{k}')\right] = 0.\end{aligned} \tag{16.82}$$

The continuum electric-field operator is then

$$\boxed{\mathbf{E}(\mathbf{r},t) = \frac{i}{(2\pi)^{3/2}}\sum_{\mu=1}^{2}\int d^3k\,\sqrt{\frac{\hbar\omega_\mathbf{k}}{2}}\,\hat{\mathbf{e}}_{\mathbf{k}\mu}a_\mu(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{h.c.}} \tag{16.83}$$

Similarly, the magnetic-field operator becomes

$$\boxed{\mathbf{B}(\mathbf{r},t) = \frac{i}{(2\pi)^{3/2}}\sum_{\mu=1}^{2}\int d^3k\,\sqrt{\frac{\hbar\omega_\mathbf{k}}{2}}\,\left(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{\mathbf{k}\mu}\right)a_\mu(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{h.c.}} \tag{16.84}$$

Lastly, the continuum field Hamiltonian is

$$\boxed{H_\text{field} = \sum_{\mu=1}^{2}\int d^3k\,\hbar\omega_\mathbf{k}a^\dagger_\mu(\mathbf{k})a_\mu(\mathbf{k}) + \frac{V}{2}\sum_{\mu=1}^{2}\int\frac{d^3k}{(2\pi)^3}\,\hbar\omega_\mathbf{k}.} \tag{16.85}$$

Notice that the vacuum energy becomes infinite for two conceptually distinct reasons. First, the total vacuum energy is proportional to the volume V, as expected for a uniform energy density, and therefore becomes infinite when the spatial volume is taken to infinity. Second, the vacuum-energy density is itself divergent because the integral includes modes with arbitrarily large wavevectors. This latter divergence is called an *ultraviolet divergence*.